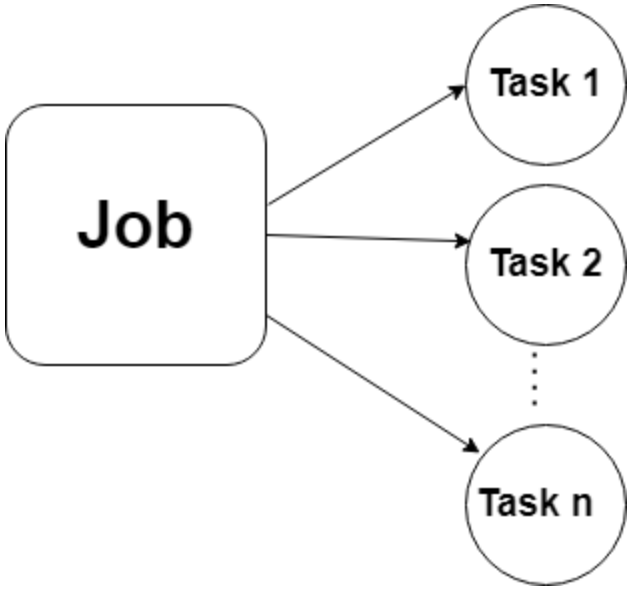
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Types of Operating System

There are several types of Operating Systems which are mentioned below.

- Batch Operating System
- Multi-Programming System
- Multi-Processing System
- Multi-Tasking Operating System
- Time-Sharing Operating System
- Distributed Operating System
- Network Operating System
- Real-Time Operating System

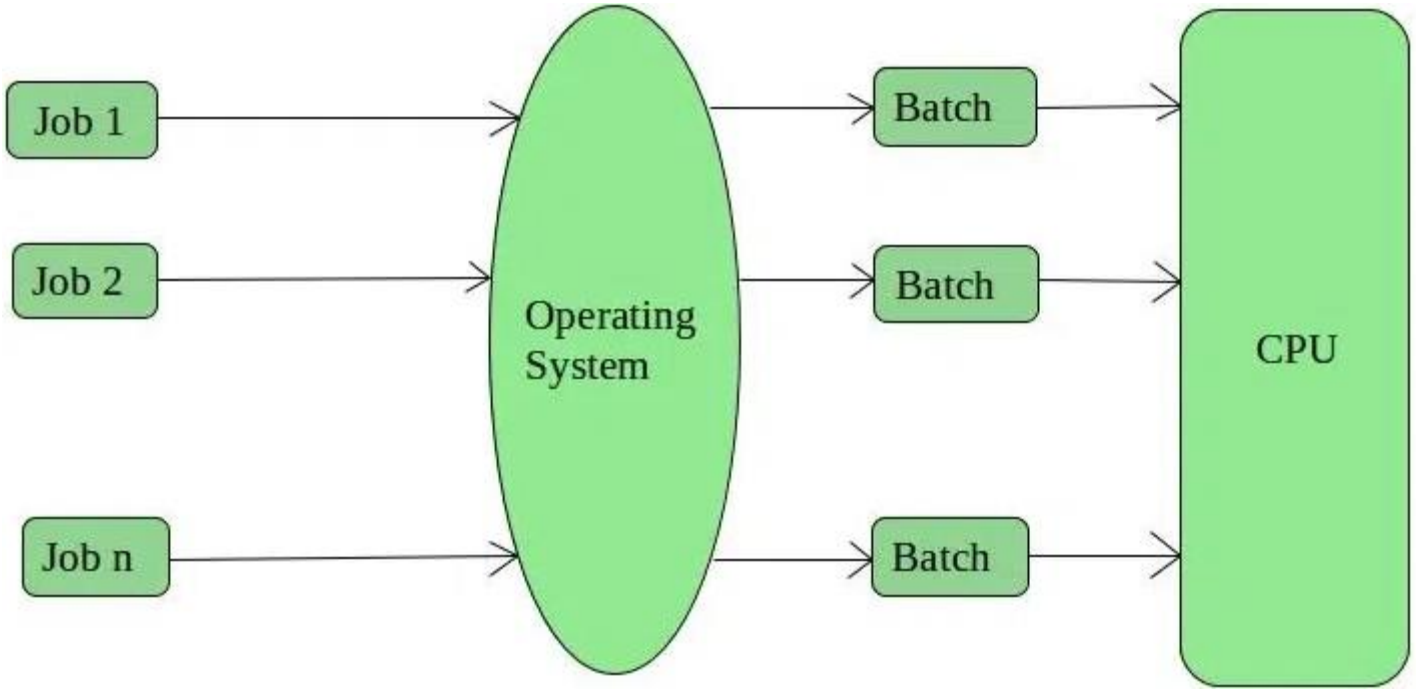
- **Job** refers to the unit of work that a computer operator or a program called a job scheduler gives to the OS to execute.



- **Uniprocessor** is a computer system type that has only one central processing unit which means that only one instruction will be executed at a time.

1. Batch Operating System

Here the operator takes similar jobs having the same requirements and groups them into batches. It is the responsibility of the operator to sort jobs with similar needs. Batch Operating System is designed to manage and execute a large number of jobs efficiently by processing them in groups.



Advantages of Batch Operating System

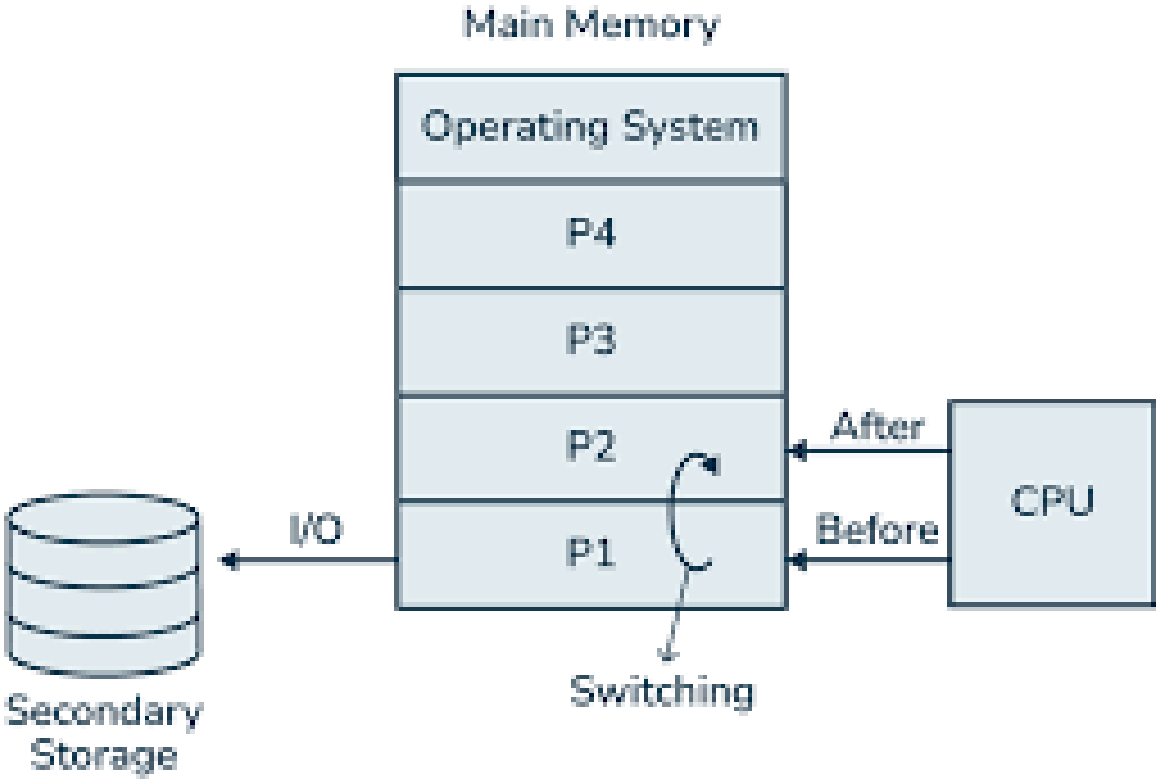
- Multiple users can share the batch systems.
- The idle time for the batch system is very less.
- It is easy to manage large work repeatedly in batch systems.

Disadvantages of Batch Operating System

- Batch systems are hard to debug.
- It is sometimes costly.
- The other jobs will have to wait for an unknown time if any job fails.
- In batch operating system the processing time high.

2. Multi-Programming Operating System

Multiprogramming Operating Systems can be simply illustrated as more than one program is present in the main memory and any one of them can be kept in execution. This is basically used for better utilization of resources.



Advantages of Multi-Programming Operating System

•Multi Programming increases the Throughput of the System.

•It helps in reducing the response time.

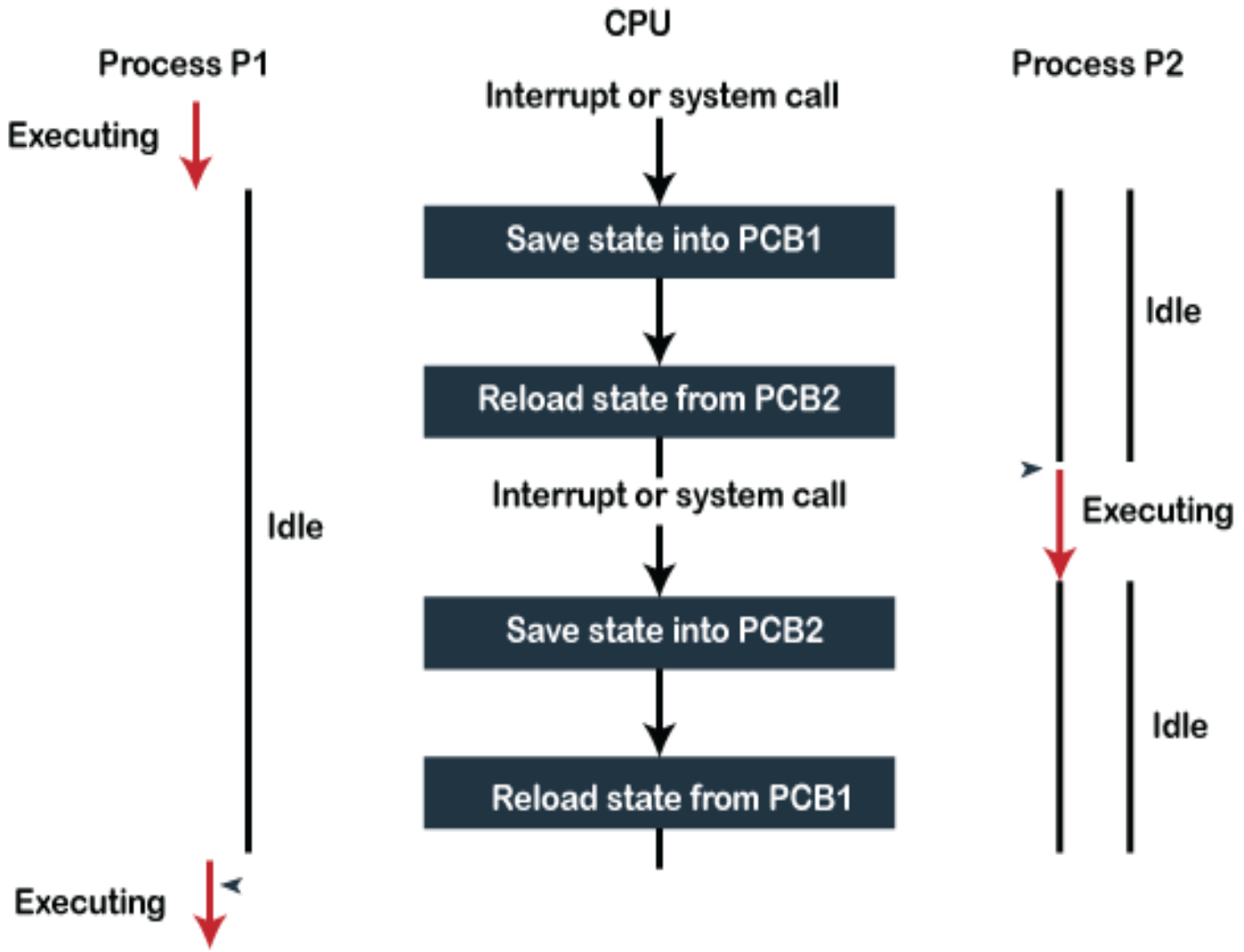
Disadvantages of Multi-Programming Operating System

•There is not any facility for user interaction of system resources with the system.

Context switching- when CPU switches to another process the system must save the state of the old process and load the saved state for the new process.

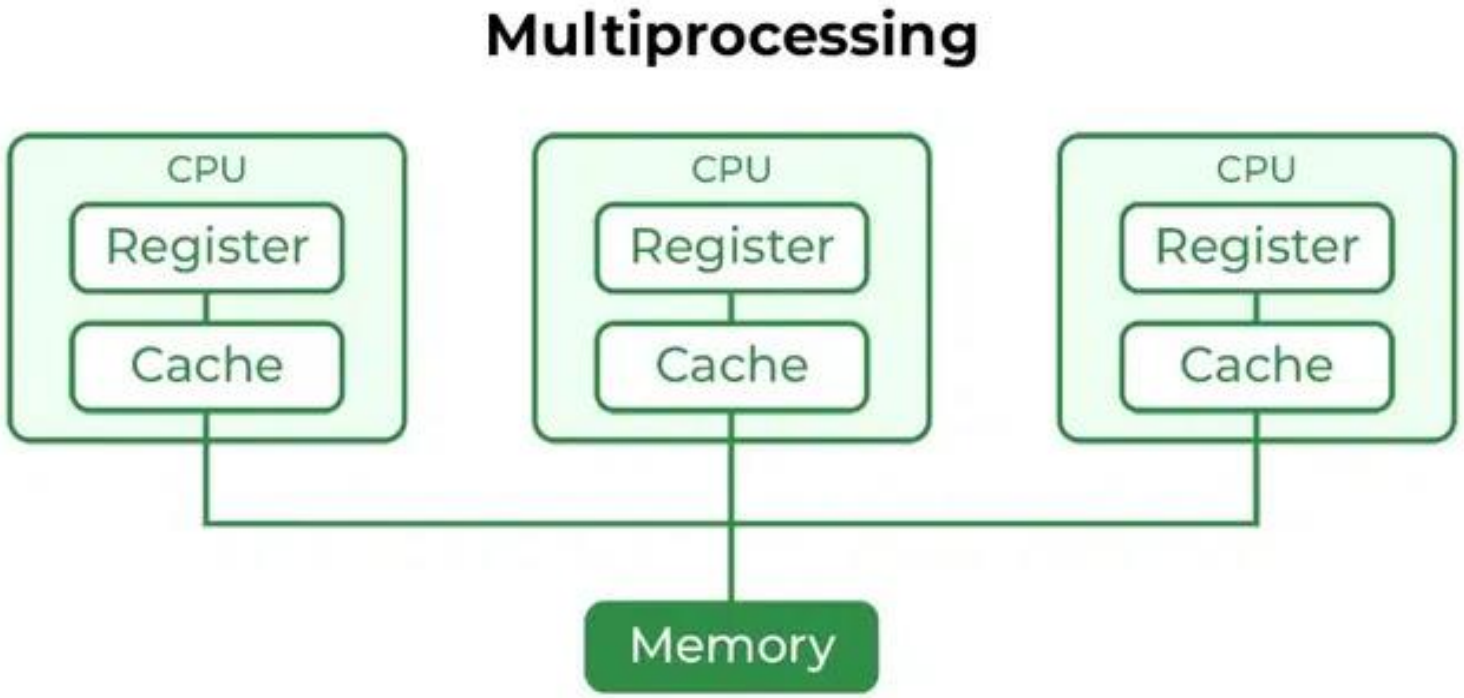
It enables all processes to share a single CPU to finish their execution and store the status of the system's tasks.

*PCB refers to **process control box** which store the context of a process


CPU
Process P1
Process P2
Executing
Interrupt or system call
Save state into PCB1
Reload state from PCB2
Idle
Interrupt or system call
Executing
Save state into PCB2
Idle
Reload state from PCB1
Idle
Executing

3. Multi-Processing Operating System

Multi-Processing Operating System is a type of Operating System in which more than one CPU is used for the execution of resources. It betters the throughput of the System.



Advantages of Multi-Processing Operating System

•It increases the throughput of the system.

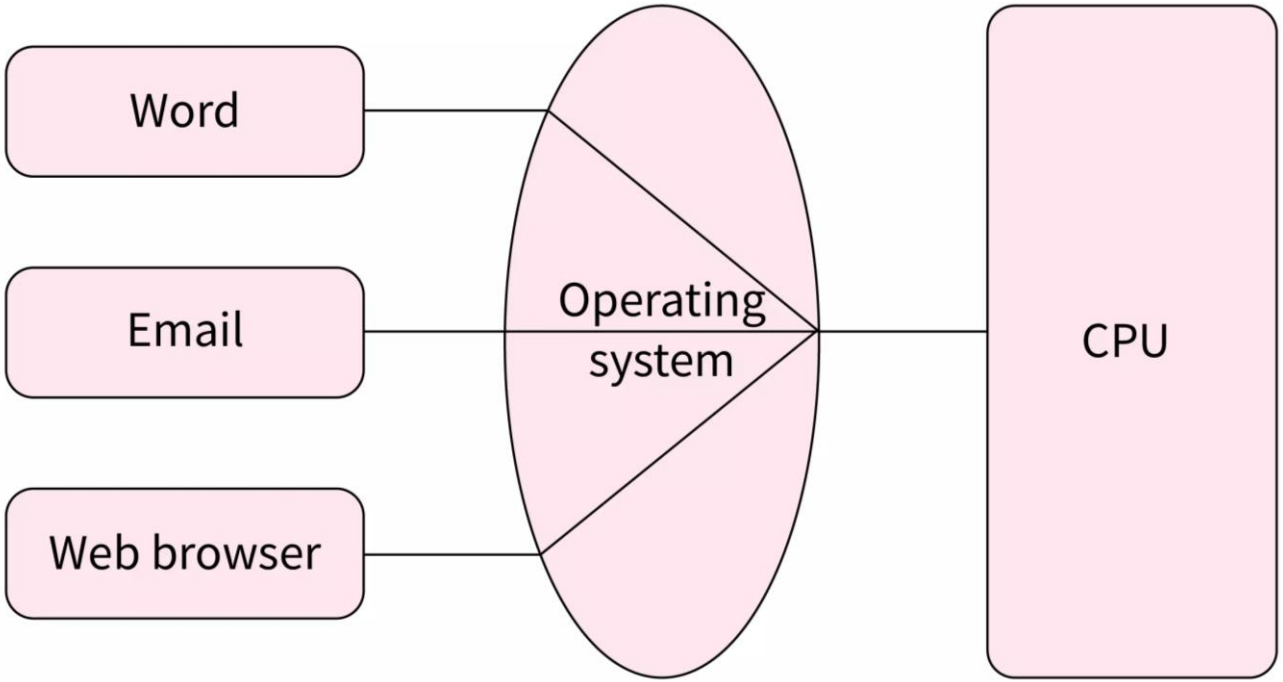
•As it has several processors, so, if one processor fails, we can proceed with another processor.

Disadvantages of Multi-Processing Operating System

•Due to the multiple CPU, it can be more complex and somehow difficult to understand.

4. Multi-Tasking/ Time sharing Operating System

Multitasking Operating System is simply a multiprogramming Operating System with having facility of a Round-Robin Scheduling Algorithm. It can run multiple programs simultaneously.



Advantages

1.task gets an equal In time sharing each process gets equal opportunity to execute because equal time quantum is given to each process.

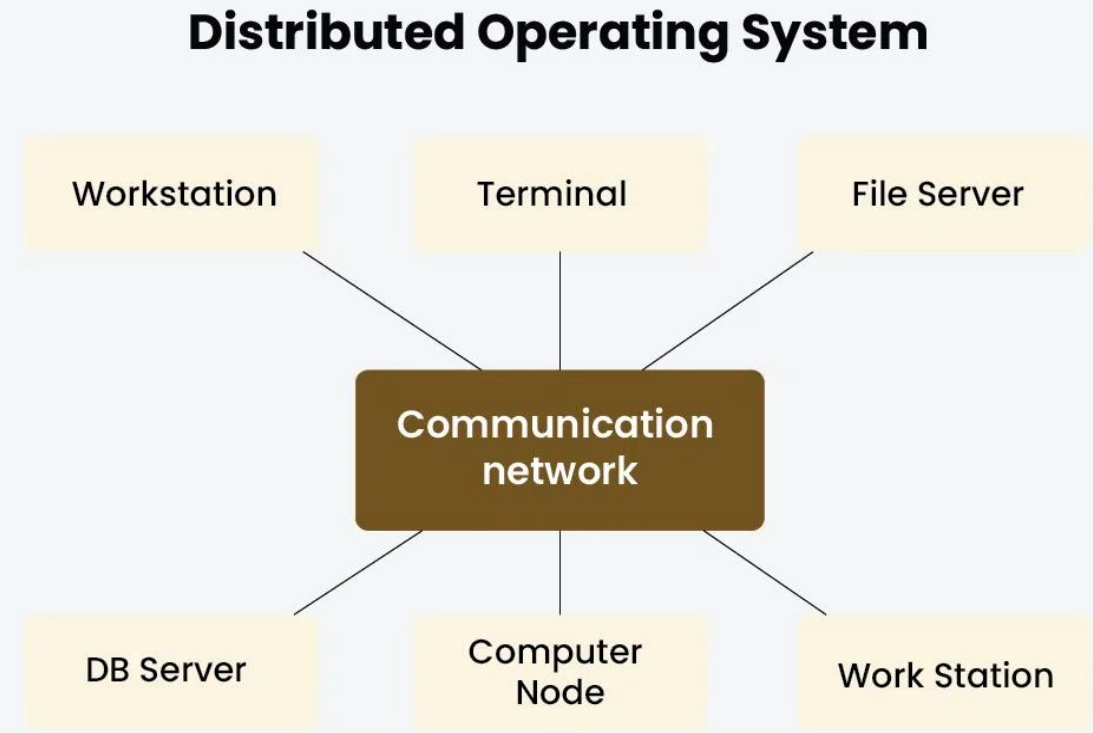
2.The CPU is always busy because of maintaining time slots, there is no wastage of CPU time.

Disadvantages

1.In the Time sharing operating system the process having higher priority does not get the chance to be executed first. This is because an equal priority is given to each process..

Distributed Operating System

In a Distributed Operating System, we have various systems and all these systems have their own CPU, main memory, secondary memory, and resources. These systems are connected to each other using a shared communication network.



Advantages :

Since the systems are connected with each other so, the failure of one system can't stop the execution of processes because other systems can do the execution.

Disadvantages:

Since the data is shared among all the computers, so to make the data secure and accessible to few computers, you need to put some extra efforts.

Embedded Operating System

An Embedded Operating System is designed to perform a specific task for a particular device which is not a computer.

Advantages:

- Since it is dedicated to a particular job, so it is fast.
- Low cost.
- These consume less memory and other resources.

Disadvantages:

Only one job can be performed.

Thank you!